IoT-Based Automatic Pill Dispenser for Medication Adherence and Remote Caregiver Alerts

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***Abstract*—Missed or incorrect medication intake is a major cause of treatment failure, especially among elderly patients and those with cognitive or motor impairments. This paper presents the design of a low-cost automatic pill dispenser based on Internet of Things (IoT) technology. The system uses an ESP32 microcontroller, a real-time clock (RTC) for accurate scheduling, and a servo-driven rotating compartment disc that releases the prescribed dose at the programmed time. A buzzer, LED and display alert the patient, while an infrared (IR) sensor confirms that the pill has been taken. If the dose is not collected within a set time, a notification is sent to the caregiver through the internet (MQTT/Wi-Fi) and, as a fallback, through SMS using a GSM module. Medication logs are stored on a cloud dashboard for doctors and family members. The proposed system is simple, affordable and easy to build with commonly available components, and aims to improve adherence, reduce caregiver burden and support independent living.**

***Index Terms*—Pill dispenser, IoT, ESP32, medication adherence, MQTT, healthcare, servo motor, caregiver alert**

I. Introduction

Medication non-adherence is a global healthcare problem. Studies show that a large share of patients with chronic diseases do not take their medicines as prescribed, leading to hospital readmissions, disease progression and higher healthcare costs [2]. The problem is more severe for elderly people, who often manage several medicines a day and may suffer from memory loss, poor eyesight or tremors that make handling small pills difficult [1].

Conventional solutions such as weekly pill organizers depend on the patient to remember the schedule and to sort the pills correctly. Smart pill caps and commercial automated dispensers solve part of the problem, but they are often expensive, require complex set-up, or do not inform the caregiver when a dose is missed [7].

The Internet of Things (IoT) offers an effective approach to this problem. By connecting a small embedded device to the internet, medication schedules can be updated remotely, intake events can be logged, and alerts can be forwarded to caregivers in real time [4].

In this work, we present an IoT-based automatic pill dispenser that:

- dispenses the correct pill automatically at the scheduled time without the user opening any container;
- alerts the patient using sound, light and a text display;
- detects whether the pill has been removed from the tray;
- notifies the caregiver through Wi-Fi (MQTT/cloud) and GSM SMS if the dose is missed; and
- stores the adherence history for later review.

The rest of the paper is organized as follows. Section II reviews related work. Section III lists the components used. Section IV describes the system architecture and circuit diagram. Section V explains the working and software flow. Section VI concludes the paper with future scope.

II. Literature Survey

Several researchers have proposed smart pill boxes, caps and dispensers. Table I summarizes the key works reviewed for this paper. From the survey, it is observed that most existing systems either lack real-world testing, depend on costly hardware, or do not provide a fallback alert to the caregiver when the internet is unavailable. The proposed system addresses these gaps by using low-cost, easily available modules and adding a GSM-based SMS backup to the Wi-Fi/MQTT alert path.

III. Components Used

Table II lists the hardware and software components of the proposed system.

IV. System Architecture and Circuit Diagram

The ESP32 is the central unit of the system. The RTC provides the current time through the I2C bus, which is shared with the LCD. When the time matches a stored schedule, the ESP32 commands the servo to rotate the disc by one compartment so that the pill falls into the collection tray. The buzzer and LED are then activated. The IR sensor monitors the tray and, once the pill is removed, the alerts are switched off and the event is logged and published to the cloud. If the pill is not removed within the allowed time, an alert is published through MQTT and an SMS is sent through the SIM800L module.

Fig. 1 shows the connection diagram and Table III gives the pin mapping.

V. Working Principle and Software Flow

A. Scheduling

The caregiver or patient enters the medicine schedule (time and compartment number) through the dashboard or a mobile application. The schedule is received by the ESP32 over

TABLE I
SUMMARY OF RELATED WORK ON SMART PILL DISPENSERS

S.No	Paper Title	Year	Key Takeaway	Advantages
1	MedDock: A 3D-Printed Smart Pill Dispenser with Sensitive Textile Sensor for Adherence Monitoring [1]	2024	3D-printed smart cap with servo-based single-pill release, ESP8266 and MQTT; textile force-resistive sensor detects pill drop and pickup; tested in homes of 8 participants for 3 days.	Real in-home evaluation; low cost; fits standard pharmacy bottles; high usability rating.
2	Smart Pill Dispenser [3]	2020	Wooden tray dispenser using a microcontroller and stepper motor with a companion Android application for reminders.	Automatic scheduling of reminders; simple mechanical design.
3	Remote Monitoring Applying IoT to Improve Control of Medication Adherence in Geriatric Patients with a Complex Treatment Regimen, Lima-Peru [4]	2020	IoT-based 3D-printed pill box with capacitive sensing and a mobile app to schedule and detect medication intake.	Remote monitoring of elderly patients; wireless protocol benchmarking.
4	Design of a Smart Medical Box for Automatic Pill Dispensing and Health Monitoring [5]	2023	Smart box with ultrasonic sensor and Raspberry Pi to detect hand movement, with fingerprint-based user authentication.	Secure access; combines dispensing with health monitoring.
5	DIALCAT: Diabetes as an Accelerator of Cognitive Impairment and Alzheimer's Disease, Comprehensive Approach and Adherence to Treatment [6]	2019	mHealth system with a smart dispenser and companion application to send reminders and monitor intake.	Improves treatment adherence for chronic patients; patient engagement through app.
6	Technologies for Medication Adherence Monitoring and Technology Assessment Criteria: Narrative Review [7]	2022	Reviews existing adherence-monitoring technologies and proposes criteria to assess them; notes cost and usability barriers.	Gives a clear framework to compare technologies and identify research gaps.

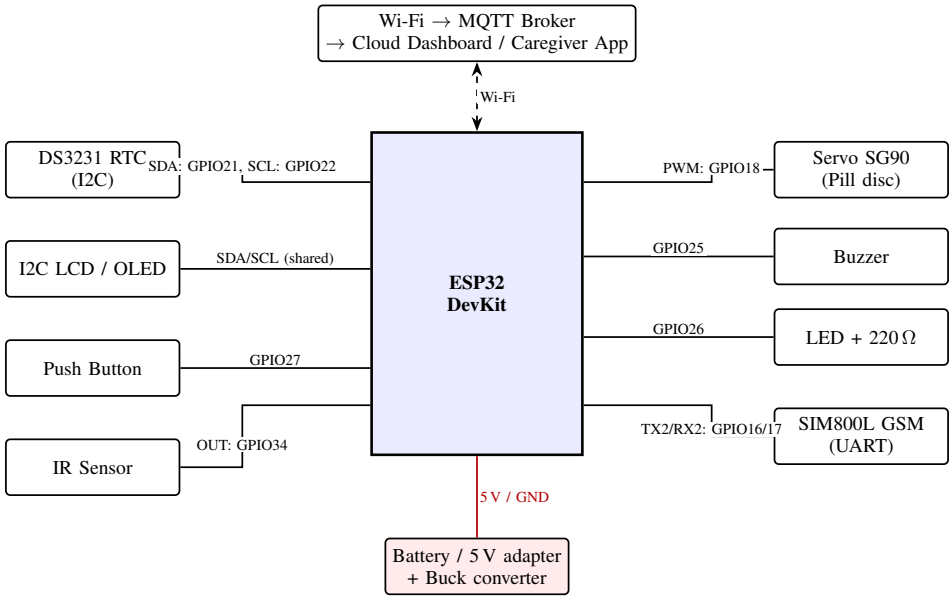


Fig. 1. Circuit (connection) diagram of the IoT-based automatic pill dispenser. All modules share a common ground. The servo and SIM800L are powered from the buck converter and not from the ESP32 pin.

MQTT and stored in its non-volatile memory, so that dispensing continues even if the internet fails.

B. Dispensing and Alerting

At each scheduled time the ESP32 compares the RTC time with the stored schedule. On a match, the servo rotates the

TABLE II
HARDWARE AND SOFTWARE COMPONENTS

S.No	Component	Purpose
1	ESP32	Main controller with built-in Wi-Fi; runs scheduling logic and MQTT client
2	Servo motor (SG90) / Stepper motor	Rotates the pill compartment disc to release one dose
3	DS3231 RTC module	Keeps accurate time and date even during power loss
4	IR sensor module	Detects presence or removal of the pill in the collection tray
5	Buzzer	Audible alert at dose time
6	LED (with 220 Ω resistor)	Visual alert and status indication
7	16×2 I2C LCD / OLED	Displays time, medicine name and messages
8	SIM800L GSM module	Sends SMS to caregiver when internet is unavailable
9	Push button	Manual acknowledgement / snooze by the patient
10	Buck converter	Steps down battery/adapter voltage to stable 5 V (and 4 V for GSM)
11	Battery cells with holder / 5 V adapter	Power supply and backup
12	Breadboard, jumper wires, 3D-printed or acrylic pill disc	Prototyping and mechanical structure
13	Arduino IDE, MQTT broker, cloud dashboard	Programming, data transfer and monitoring

TABLE III
PIN CONNECTIONS

Module	Module Pin	ESP32 Pin
DS3231 RTC	SDA / SCL	GPIO21 / GPIO22
I2C LCD	SDA / SCL	GPIO21 / GPIO22
Servo SG90	Signal	GPIO18
Buzzer	+	GPIO25
LED	Anode (via 220 Ω)	GPIO26
Push button	One terminal	GPIO27 (pull-up)
IR sensor	OUT	GPIO34
SIM800L	TX / RX	GPIO16 / GPIO17
All modules	VCC / GND	5 V (3.3 V for RTC/IR) / GND

disc to the next compartment and the pill drops into the collection tray. The buzzer, LED and LCD message "Take your medicine" are activated simultaneously.

C. Intake Detection

The IR sensor continuously monitors the tray. When the pill is removed, or the patient presses the confirmation button, the alerts are turned off, the LCD shows "Dose taken", and the timestamp is published on the MQTT topic for the dashboard.

D. Missed-Dose Notification

If the pill is not removed within a defined time window (for example 10 minutes), the system publishes a "missed dose" message to the caregiver through MQTT. If Wi-Fi is not available, the SIM800L module sends an SMS to the registered number.

E. Algorithm

1) Initialize ESP32, RTC, LCD, servo, GSM and Wi-Fi/MQTT.
2) Read current time from RTC.
3) If time equals a scheduled time, rotate servo to the next compartment.
4) Switch on buzzer, LED and display message.
5) Read IR sensor / button. If pill taken, stop alerts and log the event.
6) If timeout occurs, send missed-dose alert through MQTT and SMS.
7) Repeat from step 2.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented the design of a low-cost IoT-based automatic pill dispenser that dispenses medicines at scheduled times, alerts the patient, confirms intake and informs the caregiver when a dose is missed. The use of an ESP32, RTC, servo-driven disc, IR sensing and a GSM backup makes the system reliable and easy to build with common components. In future, the system can be extended with a camera or OCR-based medicine verification, voice reminders in regional languages, a mobile application with refill alerts, and integration with hospital records. Testing with a larger group of elderly users is also planned to validate its effectiveness.

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